

Ishan Pardhi

B.Tech - Computer Science and Engineering
Vellore Institute of Technology Bhopal

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Vellore Institute of Technology Bhopal	9.14 (Current)	2023-Present
Senior Secondary	MP Board	78.6%	2022
Secondary	CBSE Board	94.6%	2020

PROJECTS

- **Online Campus Security Management System**
Designed scalable full-stack web application using Agile methodologies [GitHub Link](#)
 - Developed a scalable web application using Node.js, Express.js, MongoDB for back-end and HTML, CSS, JavaScript for front-end, ensuring efficient data management and user authentication.
 - Implemented features like leave applications, salary calculations, and automated email notifications using RESTful APIs and database optimization techniques.
- **Brain Tumor Classification using ResNet-50**
Applied machine learning and deep learning techniques for computer vision tasks [GitHub Link](#)
 - Engineered a deep learning model with ResNet-50 architecture to classify brain tumors from MRI scans, achieving over 90% accuracy through data augmentation and transfer learning.
 - Implemented data preprocessing and augmentation on a Kaggle dataset using TensorFlow and Keras, focusing on convolutional neural networks (CNNs).
 - Trained the model for 50 epochs utilizing the Adam optimizer and categorical cross-entropy loss function, with hyperparameter tuning for optimal performance.
- **Real-Time Heart Rate Detection from Webcam**
Utilized computer vision and signal processing algorithms [GitHub Link](#)
 - Developed a computer vision system in Python using OpenCV to detect heart rate in real-time from a user's facial video stream, integrating image processing pipelines.
 - Applied digital signal processing techniques to extract photoplethysmographic (PPG) signals from pixel intensity fluctuations, employing Fourier transforms for frequency analysis.
 - Achieved a high degree of accuracy, with a low error margin of ± 5 BPM when validated against medical-grade devices, showcasing robust algorithm implementation.

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Java, JavaScript, SQL, HTML, CSS
- **Frameworks & Libraries:** React.js, Node.js, Express.js, TensorFlow, Keras, OpenCV, Scikit-learn, Mongooose
- **Databases & Tools:** MongoDB, SQL, Git, GitHub, Docker, AWS, RESTful APIs
- **Methodologies:** Agile, Scrum
- **AI:** Prompt Engineering, AI-Assisted Development, Spec-Driven Development (Spec Kit), AI Skill Creation

KEY COURSES TAKEN

- **Computer Science:** Computer Programming, Discrete Mathematical Structures, Data Base and Information Systems, Data Structures and Algorithms, Automata Theory and Logic, Logic Design, Software Engineering, Design and Analysis of Algorithms, Operating Systems, Computer Architecture, Optimization Algorithms and Techniques, Parallel Computing, Cloud Computing, Data Structures, Algorithms, Machine Learning, Deep Learning, Computer Vision.
- **Mathematics:** Calculus, Linear Algebra, Differential Equations, Complex Analysis, Numerical Methods, Probability and Statistics

COMPETITIVE PROGRAMMING

- Solved 500+ algorithmic challenges on platforms like LeetCode, Codeforces, and GeeksforGeeks, focusing on time and space complexity optimization.
- Profiles: [LeetCode](#) | [Codeforces](#) | [InterviewBit](#) | [GeeksforGeeks](#)

CERTIFICATIONS

- **Introduction to Machine Learning:** NPTEL (Elite) [↗](#)
- **MERN Full Stack:** Ethnus Certificate of Completion (Code: 3LKMCQVY) [↗](#)
- **Programming in Java:** Vityarathi [↗](#)
- **Python Essentials:** Vityarathi [↗](#)
- **Python Basics:** HackerRank [↗](#)
- **Linux Tutorial:** Great Learning Academy [↗](#)
- **The Bits and Bytes of Computer Networking:** Coursera [↗](#)
- **Fundamentals of AI and ML:** Vityarathi [↗](#)
- **MATLAB Onramp:** MathWorks [↗](#)
- **Imedita Certificate:** Imedita [↗](#)